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Spiny mouse (Acomys) stereotaxic injection V.2

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We use this protocol and it's working

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Abstract

Stereotaxic injection for spiny mice (Acomys dimidiatus)

Materials

Equipments

- Isoflurane induction chamber
- Stereotaxic frame (Kopf Instruments)
- Nose cone designed for neonatal rats (RWD #68601)
- Heating pad (for intraoperative and recovery use)
- Cup horn sonicator (if injections require fibril preparation, optional)

Anesthesia and Monitoring

- Isoflurane vaporizer and oxygen supply
- MouseSTAT Jr. pulse oximeter and heart rate monitor (Kent Scientific)

Surgical Tools and Consumables

- Autoclave-sterilized surgical scissors (preferred over scalpel for scalp incision)
- Sterile forceps and needle holders
- 10 µL Hamilton syringe (Hamilton #1701) with 33-gauge needle
- Non-absorbable nylon sutures (Ethicon #NW3353)
- Sterile surgical drapes and gloves

Reagents and Analgesics

- Isoflurane (for anesthesia)
- Ketoprofen (5 mg/kg, SC, pre-surgery)
- Buprenorphine-XR (3.25 mg/kg, SC, pre-surgery)
- Bupivacaine solution (0.25% in sterile saline; 100 µL SC at incision site)
- Chlorhexidine scrub and chlorhexidine solution (for site disinfection)
- Sterile saline (for dilution and tissue moisture maintenance)

1 Anesthetize spiny mice in an isoflurane induction chamber until loss of righting reflex is observed.

2 Transfer the animal to a stereotaxic frame (Kopf Instruments) equipped with a nose cone designed for neonatal rats (RWD #68601) to maintain anesthesia during the procedure.

3 Place the animal on a heating pad to maintain core body temperature.

4 Maintain anesthesia with isoflurane throughout the surgery, continuously monitoring oxygen saturation and heart rate using a MouseSTAT Jr. pulse oximeter and heart rate monitor (Kent Scientific). Oxygen saturation should be kept above 95%, and heart rate is typically 350–400 bpm initially, stabilizing at 250–300 bpm about 30–45 minutes after induction.

5 Immediately before surgery, administer ketoprofen (5 mg/kg, SC) and buprenorphine-XR (3.25 mg/kg, SC) for systemic analgesia.

6 Disinfect the surgical site by applying chlorhexidine scrub and chlorhexidine solution alternately three times.

7 Inject 100 µL of 0.25% bupivacaine solution (in sterile saline) subcutaneously over the skull to provide local analgesia and maintain scalp moisture.

8 Using sterile surgical scissors, make a midline incision over the skull, avoiding scalpels to minimize shearing, which is a known concern for spiny mouse tissue.

9 Perform all surgical steps under sterile conditions, using autoclave-sterilized surgical tools and aseptic technique.

10 Perform stereotaxic injections using a 10 µL Hamilton syringe (Hamilton #1701) fitted with a 33-gauge needle, according to the planned coordinates. (Injection site coordinates were chosen from published *Acomys* atlas by Vitorino et al, 2021 (DOI:10.1002/cne.25329)

11 After injections, gently release the animal from the stereotaxic ear bars before closing the incision to reduce skin tension and prevent tearing.

12 Close the incision using non-absorbable nylon sutures (Ethicon #NW3353), ensuring proper alignment of the wound edges.

13 Allow the animal to recover on a heating pad and closely monitor until fully ambulatory.

14 Continue to monitor animals regularly for the next several days for signs of pain, distress, or surgical complications, providing supportive care as needed.

Protocol references

Sayan Dutta 2026. C57 mouse stereotaxic injection. protocols.io
https://dx.doi.org/10.17504/protocols.io.14egnr4ozl5d/v1

Vitorino et al; Coronal brain atlas in stereotaxic coordinates of the African spiny mouse, Acomys cahirinus; J Comp Neurol. 2022 Aug;530(12):2215-2237. doi: 10.1002/cne.25329